Solutions and model answers for Tasks 1 to 5



Task 1 answers

1 FR

Search by cuisine

A function the system performs.

2 NFR · Performance

Results in 2 s

How fast, with a measurable target.

3 FR

SMS confirmation

Behavior triggered by an event.

4 NFR · Security

Salted passwords

How data is protected.

5 NFR · Capacity

5,000 users

A load the system must handle.

6 FR

Admin manages restaurants

An admin function.

7 NFR · Availability

99.9% uptime

Measured as uptime percentage.

8 FR

Cancel within 5 min

A business rule, so functional.

9 NFR · Usability

First order in 3 min

Testable with a user test.

10 Constraint

Java and Spring Boot

A technology limit on the design.

11 NFR · Compliance

Card security standard

An external regulatory rule.

12 Debatable

Encrypt payment data

Wording decides. 'Encrypt with AES-256' reads as an FR.

Test: can you demo it as a feature (FR), or does it say how well or within what limits (NFR)?

Task 2 answers (1 of 2)

Fast Performance	95% of restaurant searches return in 2 s or less and 99% in 5 s or less, with 5,000 concurrent users. **Verify:** Load test with JMeter or k6
Secure Security	All traffic uses TLS 1.2+. Accounts lock for 15 min after 5 failed logins. No critical or high findings in the pre-release penetration test. **Verify:** Security scan and penetration test
User-friendly Usability	In a test with 10 new users, at least 9 place an order in 3 min or less without help, and the SUS score is 75 or more. **Verify:** Usability test session

Task 2 answers (2 of 2)

Reliable

Availability

99.9% uptime per calendar month, excluding announced maintenance. Mean time to recover from failure is 30 min or less.

Verify: Monitoring and incident logs

Scalable

Scalability

Response-time targets still hold when load grows from 5,000 to 15,000 users by adding servers only, with no code changes.

Verify: Stress test with horizontal scaling

Common mistakes to point out

- Adjectives without numbers
- No load condition
- No way to test it

Task 3: model FRs

FR-01 Visitors can register and log in with email and password.

FR-02 Customers can search restaurants by name, cuisine and rating.

FR-03 Customers can add items to a cart and change quantities.

FR-04 Customers can pay by card or choose cash on delivery.

FR-05 Restaurant staff are notified of new orders and can accept or reject them.

FR-06 Staff can create, edit and mark menu items unavailable.

FR-07 Accepted orders go to a free rider, who updates delivery status.

FR-08 Customers see live order status and rider location.

FR-09 Customers can cancel within 5 minutes if preparation has not started.

FR-10 Admins can add, suspend or remove restaurants and review complaints.

Write each as “The system shall …”. FR-04 could be split for stricter atomicity.

Task 3: model NFRs

NFR-01 | Performance

95% of searches complete in 2 s or less at 5,000 concurrent users.

NFR-02 | Availability

The system is available 99.9% of each month.

NFR-03 | Security

Card data is never stored on our servers; payments use a certified gateway.

NFR-04 | Security

Passwords are stored using salted hashing.

NFR-05 | Usability

90% of first-time users order in 3 min or less without training.

NFR-06 | Compatibility

Runs on Android 10+, iOS 15+ and the latest two versions of Chrome, Safari and Firefox.

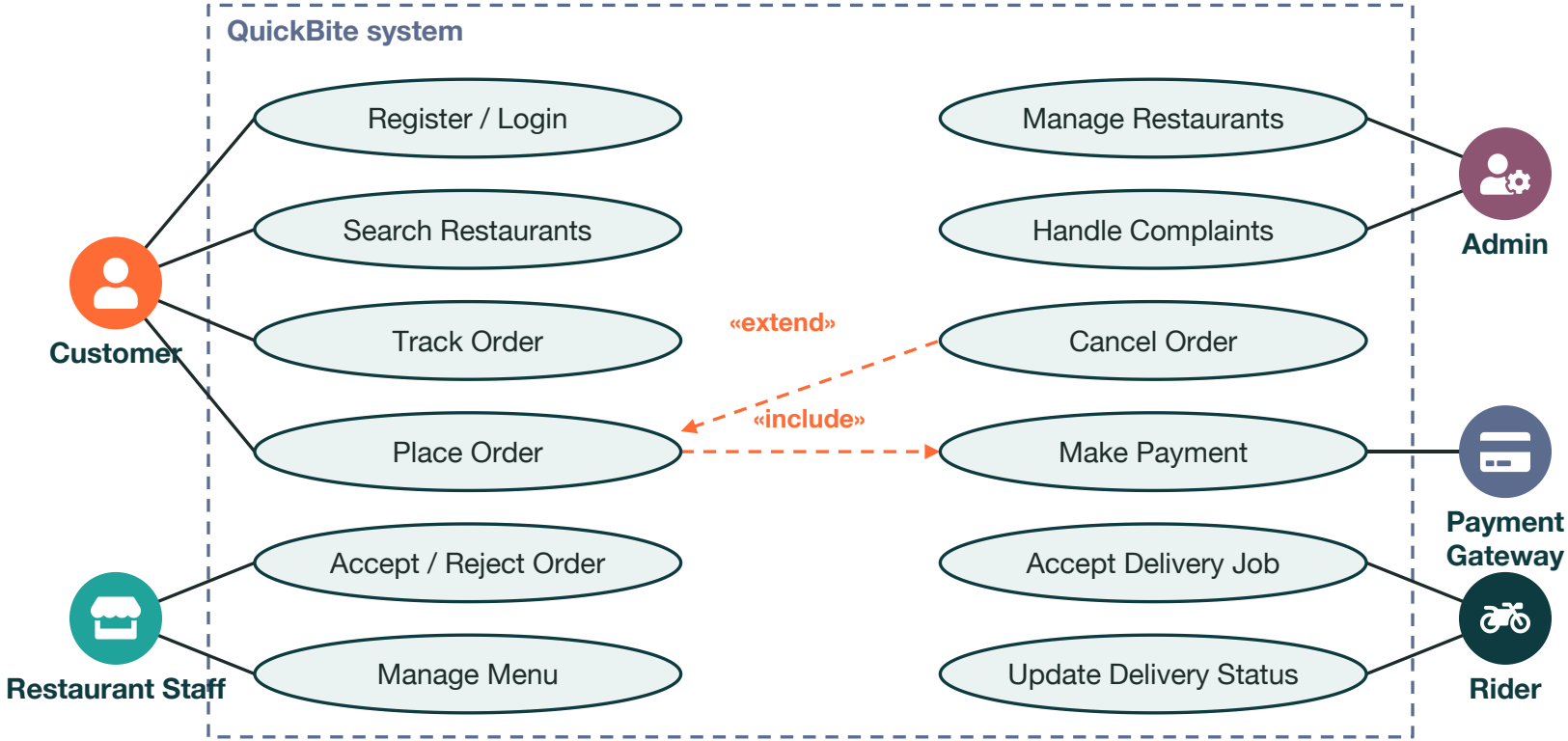
NFR-07 | Compliance

Complies with local financial regulations and data protection law.

NFR-08 | Maintainability

At least 70% unit-test coverage; modules deploy independently.

Task 4A: use case diagram



Task 4B: Place Order spec

Main flow

Actors: Customer, Payment Gateway. Trigger: Checkout.

1. Customer reviews cart and address
2. Customer selects payment method
3. System validates cart and restaurant status
4. System sends card payment to gateway
5. Gateway approves the payment
6. System creates order, notifies restaurant
7. System shows confirmation and sends SMS

Alternate flows

2a Cash on delivery: skip steps 4 and 5.
3a Item unavailable: customer edits cart.

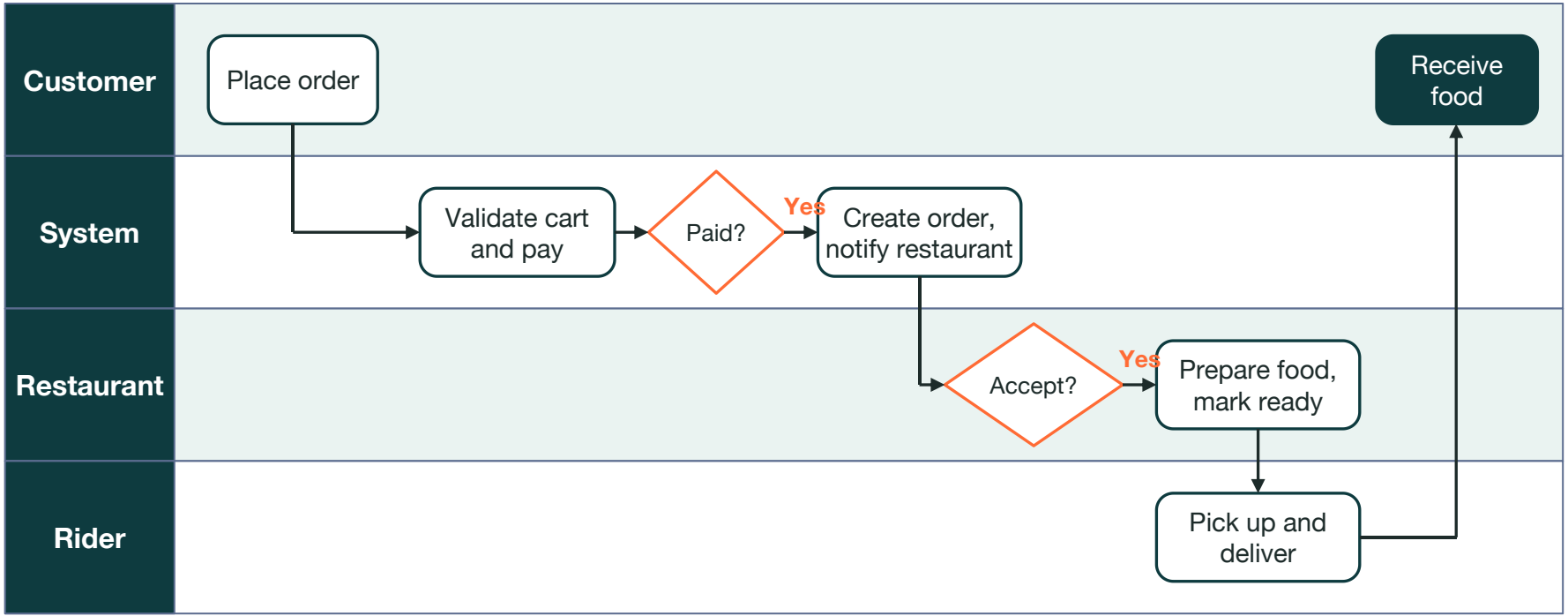
Exception flows

5a Payment declined: retry or change method.
5b Gateway timeout: no order created; retry.

Conditions

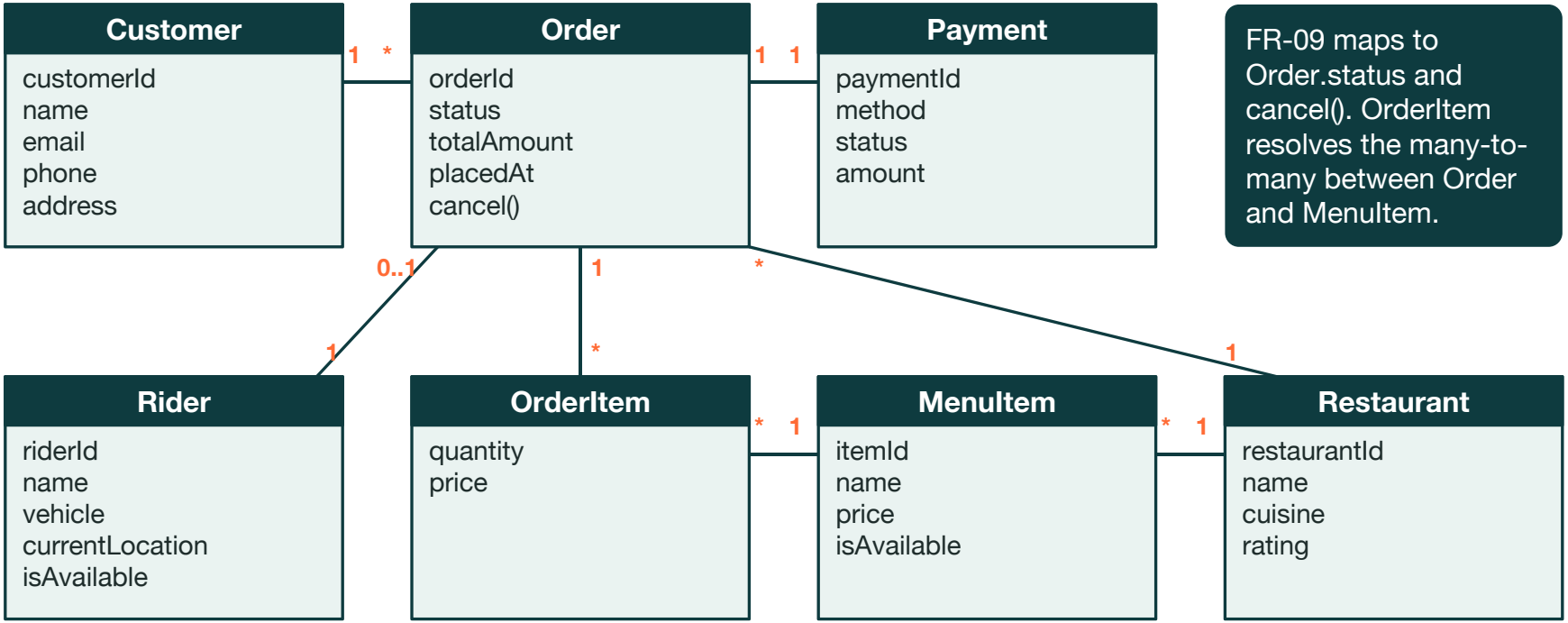
Pre: logged in, cart not empty.
Post: order saved as Placed; restaurant notified.

Task 4C: activity diagram

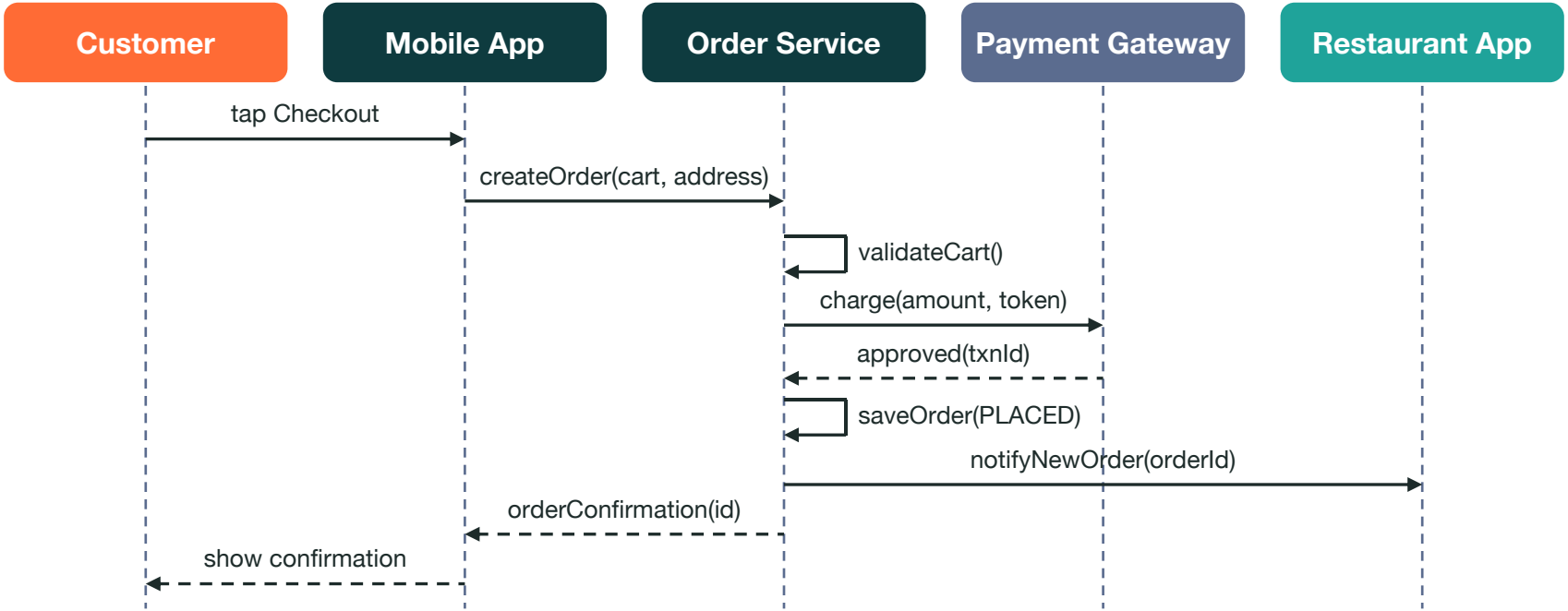


No at Paid? show error, back to checkout. **No at Accept?** refund and notify customer.

Task 4D: class diagram



Task 4E: sequence diagram



alt Payment declined: gateway returns declined, Order Service sends error(PAYMENT_FAILED) to the app.

Task 4: traceability

Requirement	Use case	Activity step	Class	Sequence message
FR-04	Make Payment	Process payment	Payment	charge()
FR-05	Accept / Reject Order	Accepts? decision	Order.status	notifyNewOrder()
NFR-03	Make Payment	n/a	Payment (stores token only)	Token sent, not card number

Every requirement should show up in at least one model

Requirement > Use case > Activity > Class > Sequence

Task 5: conflict fixes

Security vs Usability	Extra login steps slow the first order. **Fix:** guest checkout with a one-time code; 2FA only for payment changes and admin accounts.
Performance vs Cost	More servers cost more. **Fix:** autoscaling with a cost cap, search caching and a CDN for images.
Live tracking vs Privacy	Location is personal data. **Fix:** track only during active deliveries, show approximate location, delete after 30 days.
Fast release vs Maintainability	Time pressure cuts testing. **Fix:** cover payment and order modules first; 50% coverage at launch, 70% by month 9.

Task 5: MoSCoW sample

M Must

FR-01 to FR-05, FR-07, NFR-01 to NFR-04, NFR-07

S Should

FR-06, FR-08, FR-09, NFR-05, NFR-06

C Could

Advanced complaint analytics, promo codes

W Won't (this release)

Loyalty points, scheduled orders